THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE FULL SYLLABUS PAPER SET 04

Full Syllabus (Chapters 1–11) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Red rose extract (China rose) turns ___ in acids. 1

(a) green **(b)** blue
(c) yellow **(d)** dark pink or magenta

2. Which of these is acidic? 1

(a) Lime water **(b)** Soap solution
(c) Baking soda solution **(d)** Lemon juice

3. Materials through which current passes easily are: 1

(a) indicators **(b)** conductors **(c)** insulators **(d)** solvents

4. A person should not use an electric device near water because water: 1

(a) is cold **(b)** is acidic only
(c) is an insulator **(d)** can conduct electricity

5. Which non-metal conducts electricity? 1

(a) Graphite **(b)** Sulfur **(c)** Iodine **(d)** Diamond

6. The property of being drawn into wires is: 1

(a) malleability **(b)** sonority **(c)** ductility **(d)** hardness

7. The substance formed when iron rusts is: 1

(a) iron sulfate **(b)** iron chloride **(c)** iron carbonate **(d)** iron oxide

8. Substances that burn in air are called: 1

(a) neutral **(b)** combustible **(c)** inert **(d)** non-combustible

9. The menstrual cycle recurs generally every: 1

(a) 7 days **(b)** 3 months **(c)** 28–30 days **(d)** 1 year

10. Menstruation may last for about: 1

(a) 1 month **(b)** 2 weeks **(c)** 3 to 7 days **(d)** 1 day only

11. Water vapour changes into tiny droplets forming clouds by: 1

(a) melting **(b)** evaporation **(c)** infiltration **(d)** condensation

12. In summer we prefer wearing light-coloured clothes because they: 1

(a) are costly **(b)** are heavy
(c) reflect more heat **(d)** absorb more heat

13. The SI unit of speed is: 1

(a) metre **(b)** metre per second **(c)** second per metre **(d)** kilometre

14. The device that measures the distance travelled by a vehicle is the: 1

(a) stopwatch **(b)** compass **(c)** odometer **(d)** speedometer

15. The process of inhaling and exhaling air is called: 1

(a) respiration **(b)** breathing **(c)** digestion **(d)** excretion

16. Bile helps in the digestion of: 1

(a) proteins **(b)** starch **(c)** water **(d)** fats

17. The upward movement of water is aided by: 1

(a) respiration **(b)** transpiration **(c)** photosynthesis **(d)** germination

18. Which is essential for photosynthesis? 1

(a) darkness **(b)** oxygen **(c)** soil only **(d)** sunlight

19. A rainbow-coloured pattern from sunlight is due to: 1

(a) shadow **(b)** splitting of white light
(c) pinhole **(d)** reflection by plane mirror

20. Waning means the visible part is: 1

(a) constant **(b)** bigger than Sun **(c)** increasing **(d)** decreasing

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 15–25 words each.

21. Why does a factory treat its acidic waste before releasing it? 2

22. Why do we use copper in electric wires? 2

23. State the nature of oxides of metals and non-metals. 2

24. What is cyberbullying? What should one do if bullied? 2

25. Why is a metal chair cold in winter? 2

26. Name two ancient ways of measuring time. 2

27. What happens to the diaphragm during inhalation? 2

28. Why do plants kept in the dark die slowly? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. Draw the symbols (describe) of a bulb, battery and switch used in circuit diagrams. 3

30. State any two chemical changes in your body. 3

31. Give examples of radiation in daily life. 3

32. Why is bile basic and how does it help? 3

33. Why do we see our reflection in still water but not in rough water? 3

34. Why is it summer in one hemisphere and winter in the other? 3

35. Why does lemon juice taste sour while soap solution tastes bitter? 3

OR

What is the use of an indicator in a neutralisation reaction? 3

36. What would happen if all stomata were blocked? 3

OR

How do leaves exchange gases? 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 80–120 words each.

37. (i) How does a metal conduct heat? 5
(ii) Describe an activity.
(iii) Why are handles of pans made of plastic?

OR

(i) Make a table of 6 materials and test them for lustre, hardness and sound. 5
(ii) Classify as metal or non-metal.

38. **Case study:** A teenager skips meals to look thin. 5
(i) Why is it harmful?
(ii) What nutrient deficiencies occur?
(iii) What should she do?
(iv) Whom to consult?

OR

(i) Why is peer pressure common in teens? 5
(ii) Good and bad effects.
(iii) How to handle?

39. (i) How do astronomers measure time? 5
(ii) How is the day defined?
(iii) Why is the year 365 days?

OR

(i) State two uses of knowing speed in daily life. 5
(ii) Give a numerical example.
(iii) Explain why units matter.

40. (i) List the properties of light. 5
(ii) Explain two with activities.
(iii) Importance.

OR

(i) Differentiate shadow and image. 5
(ii) Differentiate plane-mirror image and pinhole image.
(iii) Which is real?

END OF QUESTION PAPER Total: 80 Marks

MASTERMIND